

“India plans to import nuclear power plants from abroad producing power worth 40,000 GWe. This will generate a large number of jobs not only in India but also in advanced countries—

Dr Om Pal Singh, visiting professor, Nuclear Engineering and Technology Programme (NETP)/Mechanical Engineering at IIT Kanpur, and former secretary, Atomic Energy Regulatory Board (AERB)



prospects. “Today, only 8000 megawatts of power is being generated but the government plans restoration of at least 25,000 megawatts of power by 2020 and 62,000 megawatts by 2032. Many more nuclear power plants are coming up and this will require much more manpower. It will definitely take time but, in a few years, there will be a huge demand for them. At present, reactors are inefficient and trained manpower is working towards increasing efficiency,” adds Dr H.C. Goel, director, Amity Institute of Nuclear Science and Technology (AINST), Noida.

Opportunities in the US

The median age in the US nuclear utility industry is nearing 50, and as much as 38 per cent of the incumbent workforce may be eligible to retire within five years. Another 10 per cent of the workforce may be lost through attrition over the same period.

Then if you factor in the expansion plans in the United States, and the even more accelerated pace of construction internationally, the bright outlook for students becomes even more apparent. The U.S. Nuclear Regulatory Commission is actively reviewing 13 applications for 22 new reactors that may be built in the United States over the next 10-20 years.

“Typically, a nuclear power plant employs 400-700 people at facilities that will operate for decades. During construction—a four- to five-year undertaking—approximately 1400-1800 people (with a peak workforce of up to 2800) are needed in jobs that include skilled trades like sheet metal work-

ers, electricians and heavy equipment operators. Worldwide, over 140 new nuclear plant projects are in the licensing and advanced planning stage, with over 60 currently under construction. The opportunities are enormous,” quips Carol Berrigan, senior director-industry infrastructure and supply chain at the US-based Nuclear Energy Institute (NEI). The NEI is the policy organisation of the nuclear energy and technologies industry and participates in both the national and global policymaking process.

Know the field

It is hard to get excited about a field that you know little about. So let’s delve a little deeper to know the work of a nuclear engineer.

Nuclear engineers develop the processes, instruments and systems for national laboratories, private industry and universities that derive benefits from nuclear energy for society. As a nuclear engineer, you may solve challenges in the areas of consumer and industrial power, space exploration, world food and water supply, environmental protection, medicine and transportation.

In other words, this branch of en-

gineering deals with the technology of harnessing energy contained in the atom generally by fission or fusion.

Where do you fit?

A nuclear power plant is a complex technological marvel that requires all kinds of engineers who will have to handle work related to power generator, reactor, fuel, radiation, electricity, materials and radioactivity.

Many nuclear engineers design, develop, monitor and operate nuclear plants used to generate electricity. They may work on the nuclear fuel cycle—production, handling and use of nuclear fuel and safe disposal of the waste produced by generation of nuclear energy. Others research the production of fusion energy. Some specialise in the development of power sources for spacecrafts that use radioactive materials. Others develop and maintain the nuclear imaging technology used to diagnose and treat medical problems.

“At present, most of the engineers who work at nuclear reactors come from engineering branches like mechanical, civil and electrical, and even include physicists and scientists who learn nuclear engineering on the job. Even from IIT Roorkee, students join Bhabha Atomic Research Centre (BARC) and get trained in nuclear energy,” notes Dr A.K. Jain, head of physics department at IIT Roorkee.

Power plants also require a large number of technicians to handle electrical things like power lines, installations and boilers. They are usually diploma holders in engineering. Nuclear engineers also carry out theoretic-

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